

Perovskite–organic tandem solar cells with a photo-transformable stabilizer

<https://doi.org/10.1038/s41586-026-10869-x>

Received: 26 September 2025

Accepted: 30 June 2026

Published online: 13 July 2026



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Wide-bandgap (WBG) mixed-halide perovskites with high bromine (Br) content, which are used as the front-cell material in perovskite–organic tandem solar cells (TSCs), often exhibit initial halide-mixing inhomogeneity and light-induced halide segregation^{1–3}, limiting the performance of perovskite–organic TSCs. Here we introduce a photo-transformable additive, 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl] benzylammonium salt (TDB), into the WBG perovskite precursor solution to establish a two-stage strategy for stabilizing the mixed-halide phase. During crystallization, TDB improves the initial halide homogeneity by suppressing the rapid precipitation of the Br-rich phase and accelerating halide mixing upon annealing. During operational illumination, TDB undergoes transformation to form a new species with stronger adsorption on the perovskite grain-boundary surfaces, which inhibits the formation of iodide-related defects and suppresses defect-assisted carrier trapping and ion migration, thereby mitigating light-induced halide segregation^{4–6}. The representative WBG perovskite (bandgap energy (E_g) = 1.88 eV) solar cell had a power conversion efficiency (PCE) of 20.01%, with an open-circuit voltage of 1.42 V, a fill factor of 85.13% and improved stability under illumination. By integrating the WBG perovskite solar cell into a monolithic perovskite–organic TSC, we achieved a PCE of 28.80%, with a certified steady-state PCE of 28.04%. The perovskite–organic TSC retained 90% of its initial PCE after 625 h of operation under the ISOS-L-1 protocol.

Perovskite–organic TSCs reduce thermalization losses by using a WBG perovskite front cell to absorb ultraviolet and visible light, together with a narrow-bandgap (NBG) organic rear cell to absorb near-infrared light^{7–9}. As the key front cell, WBG perovskite solar cells (pero-SCs) have an essential role in filtering harmful ultraviolet radiation and providing high open-circuit voltage (V_{oc}) for TSCs⁷. However, because state-of-the-art organic solar cells (OSCs) have a bandgap of around 1.30 eV, WBG perovskites that are suitable for perovskite–organic TSCs usually incorporate a higher Br content (nearly 50 mol%) to achieve a larger bandgap ($E_g > 1.85$ eV) than those in perovskite–silicon ($E_g \approx 1.7$ eV) or perovskite–perovskite ($E_g \approx 1.8$ eV) TSCs¹⁰. Consequently, this WBG perovskite front cell is more susceptible to halide separation and associated energy losses^{11–15}.

In general, the occurrence of halide separation in high-Br-content WBG perovskites is due to two interrelated stages. First (stage I), during crystallization, notable differences arise in the solubility and crystallization kinetics of PbI_2 and PbBr_2 precursors¹⁶. This dynamic mismatch makes it difficult to achieve homogeneous halide mixing^{13,16}, resulting in the formation of iodine (I)-rich and Br-rich domains. This initial halide inhomogeneity can serve as a ‘seed’ for subsequent halide segregation. Second (stage II), under illumination, holes preferentially

accumulate at the low-bandgap I-rich domains, which thermodynamically stabilizes the de-mixed state^{4,5,17}. Meanwhile, the defects reduce the activation barrier for halide-ion migration and thereby kinetically accelerate the halide segregation^{3,6,18}, which, in turn, induces more halide defects and mobile ions. Under device operation, these mobile ions lead to ion-induced electric-field screening, thereby reducing charge-collection efficiency and accelerating device degradation¹⁹.

Efforts have been made to suppress halide segregation in WBG perovskites. A cyanate-based pseudo-triple-halide alloy can induce lattice distortion and uniform iodide–bromide distribution in a 1.93-eV WBG perovskite¹². Organic compounds have been introduced to regulate the redox potential to reduce iodine and oxidize metallic lead¹⁵. However, previous studies have mostly investigated individual aspects of halide segregation. Therefore, strategies are required to reduce the initial halide-mixing inhomogeneity in stage I and suppress defect-assisted halide-ion migration under illumination in stage II.

Here we introduce TDB into the WBG perovskite precursor solution. Its photo-transformable nature enables stage-adaptive functionality, allowing TDB to sequentially address the challenges arising at two different stages in WBG mixed-halide perovskites. During stage I, the three-membered diazirine moiety of TDB engages in multisite

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interactions with lead halide and formamidinium iodide, suppressing the rapid precipitation of the Br-rich phase in the wet film. The homogeneous mixed-halide phase in the wet film provides a foundation for halide mixing during annealing, improving halide homogeneity in the final film. Consistent with these improvements, a single-junction WBG pero-SC with a bandgap of 1.88 eV achieved a PCE of 20.01%, with an increased V_{oc} of 1.42 V and a fill factor (FF) of 85.13%. During stage II, photo-transformable TDB that has accumulated at the grain boundaries is activated by the ultraviolet component of the solar spectrum and transforms into a new species, 4-(trifluoroacetyl)benzylammonium salt (TAB). TAB, with a higher dipole moment, exhibits stronger adsorption on perovskite grain-boundary surfaces and increases the formation energy of iodide-related defects, thereby suppressing defect-assisted carrier trapping and halide-ion migration and mitigating light-induced halide segregation in WBG perovskite. When paired with a NBG OSC based on PM6:BTPSe-Ph4F:MO-IDIC-2F as the rear cell, the representative perovskite-organic TSC reached a PCE of 28.80% (certified steady-state PCE of 28.04%) with a V_{oc} of 2.20 V. It retained 90% of its initial PCE after 625 h of maximum power point (MPP) tracking under the ISOS-L-1 protocol.

Role of TDB during crystallization

To investigate how the crystallization process influences phase separation in WBG perovskites that contain 50 mol% Br, we performed *in situ* grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements. Pronounced phase separation was already evident in the wet-film state after spin-coating was completed, and persisted even after annealing (Supplementary Fig. 1). On the basis of these results, we focused on the (i) spin-coating and (ii) annealing processes to track phase evolution, using the untreated precursor solution ($\text{FA}_{0.8}\text{Cs}_{0.2}\text{Pb}(\text{I}_{0.5}\text{Br}_{0.5})_3$) as the control for crystallization analysis.

First, during spin-coating (Fig. 1a), the wet film of the control sample exhibited a single diffraction peak after antisolvent dripping without a detectable δ phase or solvent-associated intermediate phase. Once spinning stopped, this single peak split immediately into two distinct peaks (Fig. 1b), indicating clear phase segregation. A similar transition was observed in *in situ* photoluminescence measurements (Supplementary Fig. 2). We infer that rapid solvent evaporation during spin-coating drives the concurrent precipitation of both Br-rich and I-rich precursors, thereby diminishing the influence of their intrinsic solubility disparity. Once spinning stops, the solubility disparity dominates the precipitation kinetics, promoting the preferential precipitation of the less soluble Br-rich phase^{6,18} (Supplementary Fig. 3). Second, during annealing (Fig. 1c), the phase separation formed in the wet film was retained at the initial stage of annealing. The diffraction peak of the control film broadened, indicating the onset of halide mixing, and then gradually narrowed with prolonged annealing^{20,21}. Nevertheless, the diffraction peak of the control sample remained noticeably broadened (Fig. 1d) rather than evolving into a sharp single peak, indicating incomplete halide homogenization^{20,21}.

To improve the initial halide homogeneity of the WBG perovskite, we introduced TDB (Fig. 1e) to regulate crystallization. Consisting of a benzylammonium core and a para-substituted trifluoromethyl diazine moiety, TDB features a three-membered diazine ring containing two electronegative N atoms (Supplementary Fig. 4). The two N atoms bearing lone-pair electrons not only serve as coordination sites for Pb cations, but also provide two hydrogen-bond acceptor sites for the N–H groups of FA^+ . As shown by the ^1H nuclear magnetic resonance (NMR) spectra (Fig. 1f), after the addition of PbI_2 or PbBr_2 , the aromatic protons ortho to the trifluoromethyl diazine group shifted upfield, consistent with the coordination between the diazine N atoms and the Pb cations, which reorients the diazine ring and alters the local electronic environment (Supplementary Fig. 5). We also investigated the role of diazine N atoms in facilitating the interactions between TDB and FA^+ .

The N–H proton signals of FA^+ exhibited splitting after the addition of TDB (Fig. 1g), indicative of hydrogen-bonding interactions between FA^+ and TDB. For ^{19}F NMR spectra, see Supplementary Fig. 5. With the ability to interact with both the inorganic and the organic components of the perovskite precursors, TDB was expected to influence crystallization and thereby regulate halide mixing during film formation.

To investigate how TDB influences the crystallization of the high-Br-content WBG perovskite, we introduced TDB into the precursor solution and monitored the crystallization process by *in situ* GIWAXS. First, during spin-coating (Fig. 1a), the TDB-treated wet film showed only slight peak broadening after spinning stopped, and the diffraction feature could be assigned mainly to a mixed-halide phase, with only minor peak variation (Fig. 1b). This suggests that TDB stabilizes the mixed-halide phase and suppresses Br-rich phase precipitation through its interactions with both lead halides and FA^+ . This wet-film state with homogeneous phase distribution provides a foundation for halide mixing during annealing.

Second, during annealing, the TDB-treated film showed faster halide-mixing kinetics than the control did, and ultimately evolved into a single diffraction peak (Fig. 1c,d). This improvement is mainly due to the homogeneous phase distribution established in the wet-film state. Meanwhile, additive-assisted grain coarsening further promotes halide mixing during thermal treatment²². To investigate this process further, we performed ^{207}Pb solid-state NMR experiments on a fully separated MAPbI_3 and MAPbBr_3 powder mixture^{22,23} (see Methods for experimental details). ^{207}Pb solid-state NMR is sensitive to the local coordination environment of Pb centres in $[\text{PbX}_6]$ octahedra. During halide mixing, the progressive exchange between I^- and Br^- continuously alters the local Pb–X configuration towards mixed $[\text{PbI}_{6-x}\text{Br}_x]$ octahedra, leading to characteristic changes in the ^{207}Pb chemical shifts. Therefore, the evolution of the ^{207}Pb NMR spectra directly reflects the evolution from separated I-rich and Br-rich phases to a mixed-halide state during thermal treatment. Compared with the control, the TDB-treated sample evolved more quickly towards a mixed-halide chemical environment (Fig. 1h), indicating accelerated thermally driven halide mixing. This effect might originate from the ability of TDB to facilitate ion transport^{22,23}. The calculated binding energies between TDB and halide ions are summarized in Supplementary Table 1. Consistent with this result, after the same duration of thermal treatment, the TDB-treated powder sample exhibited a colour closer to that of the mixed-halide phase, whereas the control sample still retained an orange hue, indicating residual Br-rich domains (Supplementary Fig. 7).

To evaluate the effect of TDB on the crystalline quality of perovskite films, we performed X-ray diffraction (XRD) measurements (Supplementary Fig. 8). With increasing TDB concentration, the diffraction peaks evolved from a random orientation to a pronounced (100) preferred orientation. GIWAXS patterns further revealed an out-of-plane preferred orientation in the TDB-treated film (Fig. 1i). To understand the development of orientational ordering, we tracked the azimuthal angular intensity of the perovskite (100) diffraction signal (around $q = 1 \text{ \AA}^{-1}$) during annealing (Supplementary Fig. 9), which indicated that TDB promotes crystallite reorientation and orientational ordering during thermal annealing²². Moreover, no diffraction signals associated with 2D perovskite were observed in the TDB-treated films. Together with the results from time-of-flight secondary ion mass spectrometry (ToF-SIMS) analysis (Supplementary Fig. 10), these observations suggest that TDB is localized predominantly at grain boundaries throughout the perovskite bulk.

The morphology of the perovskite films was investigated by scanning electron microscopy (SEM) and atomic force microscopy (AFM). Compared with the control film, the TDB-treated film had larger grains with a narrower size distribution (Supplementary Fig. 11), which suggests that TDB induces an increase in ion transport across grain boundaries, thus enabling more effective grain coarsening during annealing²².

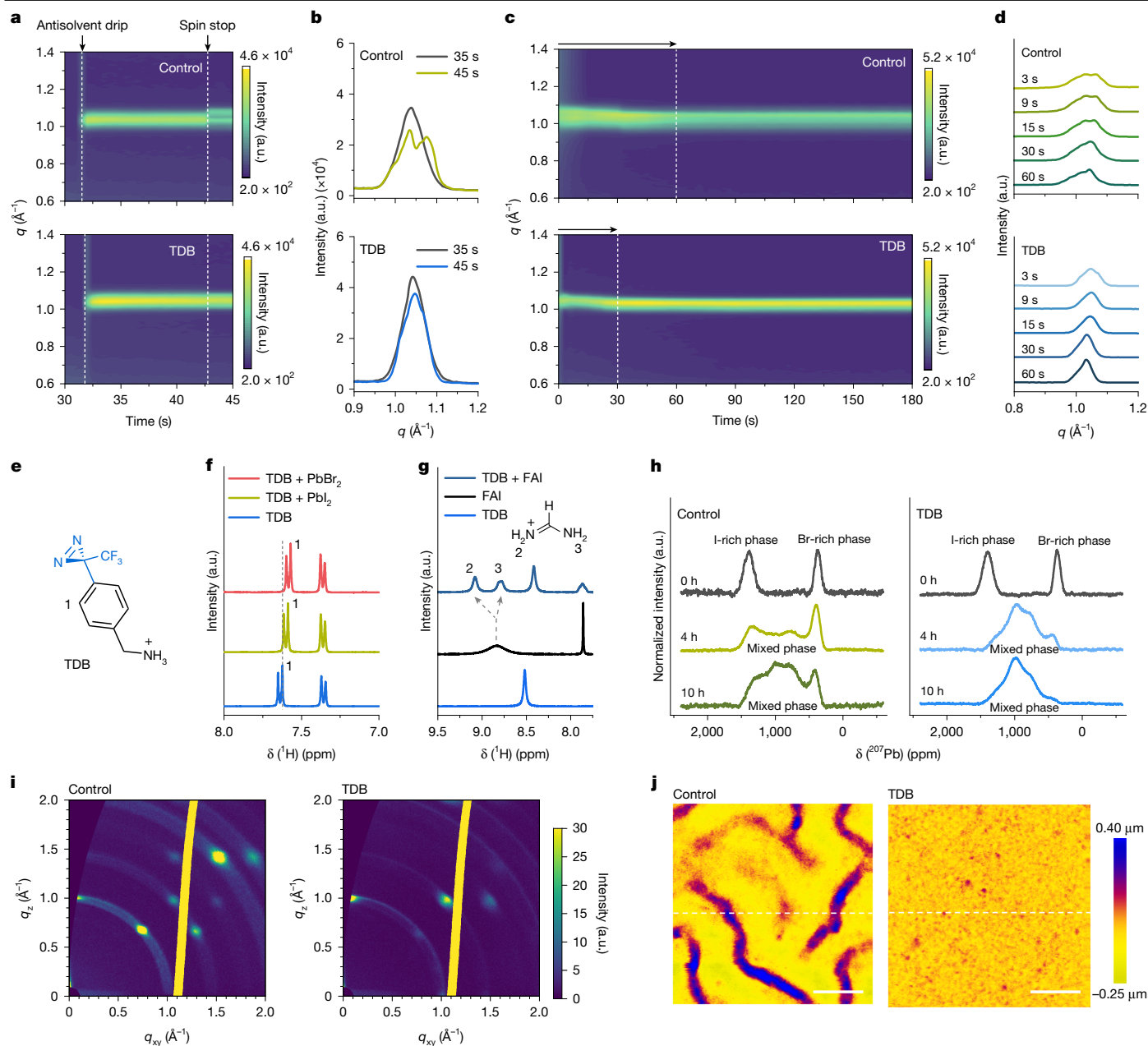


Fig. 1 | TDB improves initial halide mixing and crystal quality in WBG perovskites. **a**, In situ GIWAXS patterns recorded during spin-coating of perovskite (FA_{0.8}CS_{0.2}Pb(I_{0.5}Br_{0.5})₃) wet films without (control) and with TDB treatment. Dashed lines mark the time points at which the antisolvent was dripped (31 s) and spin-coating was terminated (43 s). **b**, Radial profiles extracted at 35 s and 45 s from the GIWAXS reciprocal-space maps of wet perovskite films without and with TDB treatment during spin-coating, highlighting the perovskite (100) diffraction signal measured at an incident angle of 1°. The profiles at 35 s and 45 s correspond to features during spin-coating and after spin-coating was stopped, respectively. **c**, In situ GIWAXS patterns recorded during annealing of the perovskite films without and with TDB treatment. The broadened peak at the early stage of annealing indicates the coexistence of mixed-halide phases with different Br and I contents, giving rise to local variations in unit-cell size. The shift of the diffraction peak towards lower q values reflects thermal dissociation of the solvent complex and subsequent

Ostwald ripening, accompanied by progressive iodide incorporation, which leads to an increase in lattice distances^{20,21}. The white dashed line marks the time point at which the film completed the initial halide mixing during thermal treatment, after which the diffraction peak shows no obvious change. **d**, Radial profiles extracted at 3, 9, 15, 30 and 60 s during annealing. **e**, Molecular structure of TDB. **f**, ¹H NMR spectra of TDB, TDB + PbI₂ and TDB + PbBr₂. **g**, ¹H NMR spectra of TDB, formamidinium iodide (FAI) and TDB + FAI. **h**, ²⁰⁷Pb NMR spectra of a physical 0.55:0.45 mixture of MAPbI₃:MAPbBr₃ powders without and with TDB, measured after mixing in the dark at 80 °C for 0 h, 4 h and 10 h. **i**, GIWAXS reciprocal-space maps of control and TDB-treated perovskite films after annealing. **j**, AFM topography images of control and TDB-treated perovskite films. Scale bars, 20 μm. The root-mean-square roughness was calculated as 86.5 nm for the control perovskite film and 26.6 nm for the TDB-treated perovskite film. The dashed line indicates the position at which the surface altitude deviation was measured.

Consistent with this, AFM revealed that the TDB-treated WBG perovskite film has a markedly smoother surface than does the control film, which showed pronounced surface wrinkles (Fig. 1j). These wrinkles are likely to be a result of compressive stress arising from slow halide mixing

during annealing, during which the initially formed Br-rich and I-rich regions remain compositionally mismatched before evolving toward a mixed-halide phase^{24,25}. Such wrinkled morphology is associated with a higher defect density²⁶ and is also unfavourable for solution-processed

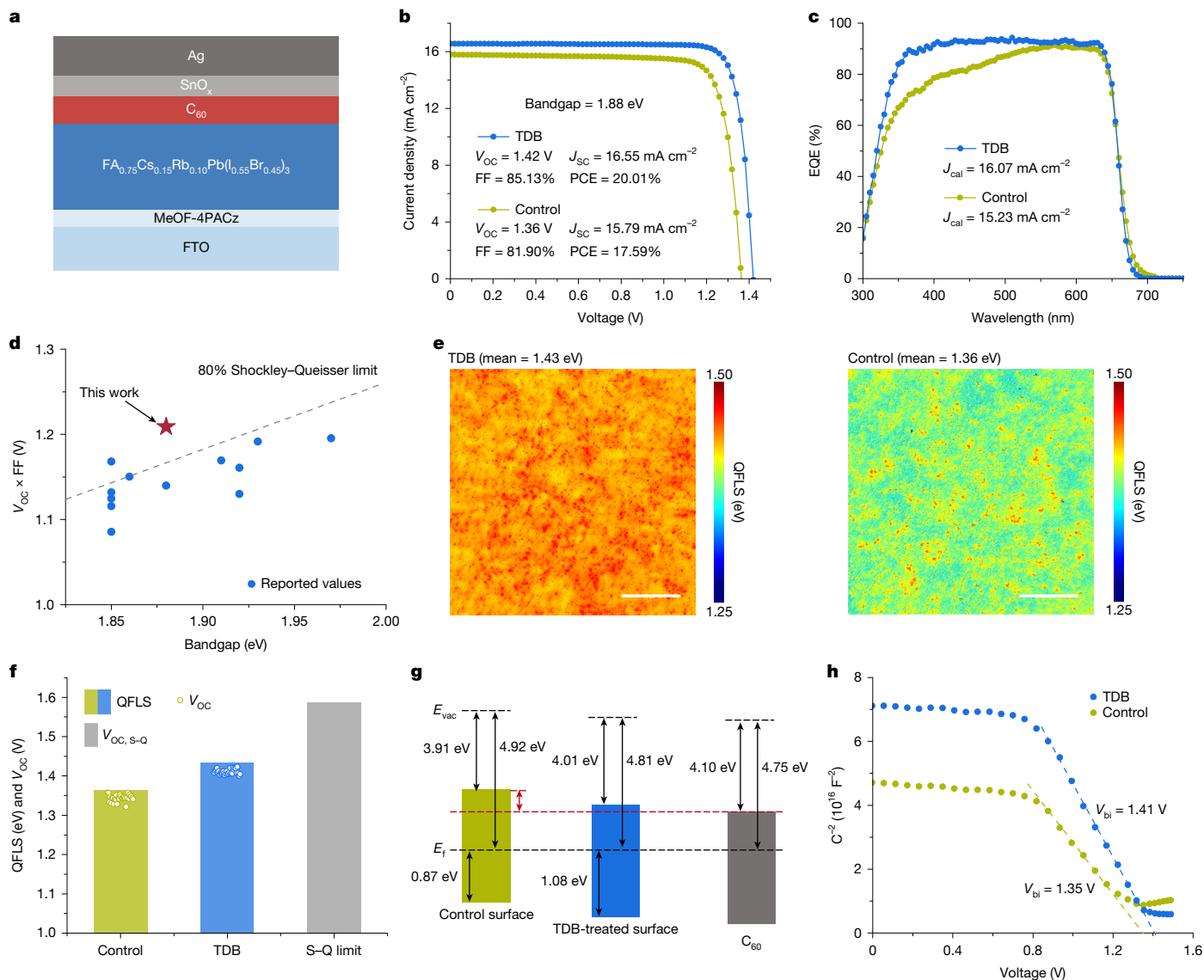


Fig. 2 | Optoelectronic properties and photovoltaic performance of WBG perovskite solar cells. **a**, Structure of single-junction WBG perovskite solar cells. **b**, J - V curves of representative TDB-treated and control WBG perovskite solar cells (active area = 0.092 cm²). **c**, EQE spectra of TDB-treated and control WBG perovskite solar cells. J_{cal} , the calculated short-circuit current density obtained by integrating the EQE spectrum over the AM1.5G solar spectrum. **d**, Comparison of the $V_{\text{oc}} \times \text{FF}$ product of this work with those of previously reported p - i - n WBG perovskite solar cells with bandgaps exceeding 1.85 eV. **e**, Spatial maps of QFLS for control and TDB-treated films.

Scale bars, 50 μm . **f**, Comparison of the V_{oc} extracted from J - V curves and corresponding QFLS values obtained from representative layer stacks, together with the S-Q limit. **g**, Band alignment at the interface between control or TDB-treated perovskite film and the electron transport layer C₆₀. The vacuum level (E_{vac}) and Fermi level (E_{f}) are indicated with dashed lines. The red dashed line denotes the LUMO level of C₆₀. **h**, Mott-Schottky plots of TDB-treated and control devices.

multi-junction devices, especially when integrated with thin (around 120-nm) organic layers. The height distributions of the control and TDB-treated films are compared further in Supplementary Fig. 12.

Optoelectronic properties and photovoltaics

On the basis of the homogeneous halide mixing and improved crystal quality of the high-Br-content WBG perovskite, we fabricated single-junction p - i - n structured (Fig. 2a) WBG perovskite solar cells ($E_{\text{g}} = 1.88$ eV) with the addition of TDB (detailed fabrication methods are provided in the Methods, the optimization of TDB concentration is shown in Supplementary Fig. 13 and the determination of E_{g} is in Supplementary Fig. 14). Figure 2b shows current density-voltage (J - V) curves of the TDB-treated and control perovskite solar cells, and the photovoltaic performance parameters are summarized in Supplementary Fig. 15 for comparison.

The representative TDB-treated WBG perovskite solar cell had a PCE of 20.01%, with a V_{oc} of 1.42 V and an FF of 85.13%. In comparison, the control perovskite solar cells showed a lower V_{oc} (1.36 V) and PCE (17.59%). The external quantum efficiency (EQE) spectra show an integrated short-circuit current density (J_{sc}) value of 16.07 mA cm⁻² for the TDB-treated perovskite solar cells—higher than the value for the control devices (Fig. 2c). Figure 2d compares the $V_{\text{oc}} \times \text{FF}$ product of our work with those of previously reported p - i - n structured WBG perovskite solar cells with a bandgap exceeding 1.85 eV (Supplementary Table 2). The TDB-treated device achieved a $V_{\text{oc}} \times \text{FF}$ product corresponding to 82.9% of the Shockley-Queisser (S-Q) limit. The V_{oc} of 1.42 V corresponds to 90% of the S-Q limit for the WBG perovskite solar cells with an E_{g} of 1.88 eV (Supplementary Fig. 17).

To examine the carrier dynamics in the WBG perovskite, we performed confocal photoluminescence imaging (Supplementary Fig. 18), which revealed a more spatially uniform emission in the TDB-treated

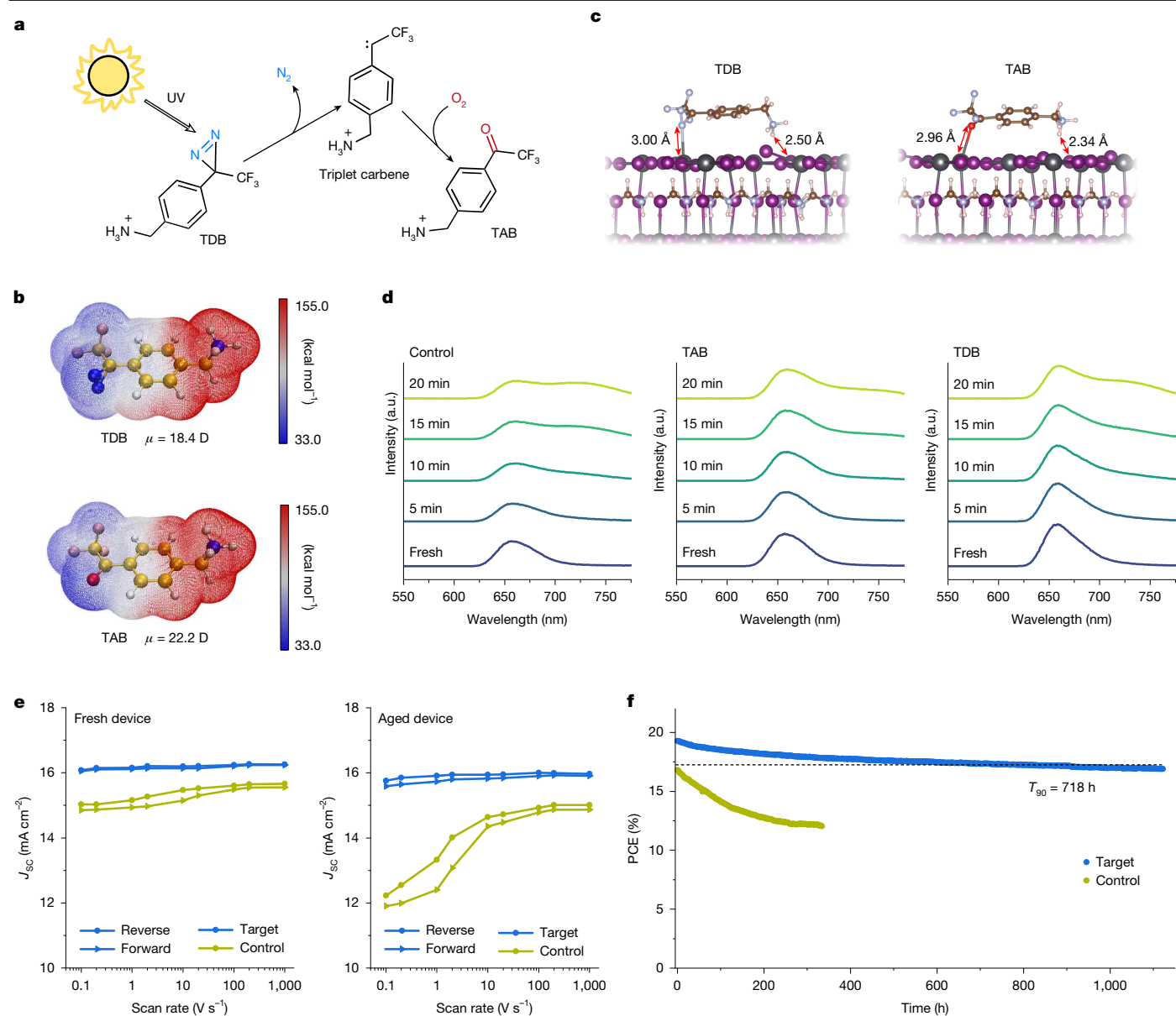


Fig. 3 | Photochemical transformation and suppression of phase separation.

a, Photo-induced reaction of TDB. **b**, Electrostatic potential surfaces of TDB and TAB. **c**, Calculated adsorption models of TDB and TAB on the perovskite surface. For TDB, the marked distances of 3.00 Å and 2.50 Å correspond to the N-Pb and H-I distances. For TAB, the marked distances of 2.96 Å and 2.34 Å correspond to the O-Pb and H-I distances. The shorter O-Pb and H-I distances indicate that TAB makes closer contact with the perovskite surface than TDB does. **d**, Photoluminescence evolution for control, TAB-treated and

film, compared with the control film, suggesting a more homogeneous halide distribution. The enhanced photoluminescence emission and prolonged carrier lifetimes extracted from time-resolved photoluminescence measurements (Supplementary Fig. 19 and Supplementary Table 3) also suggested reduced deep-level trap formation.

Furthermore, quasi-Fermi level splitting (QFLS) data were extracted from HTL/perovskite/*cis*-CyDAI₂/ETL stacks (Fig. 2e). The QFLS distributions are shown in Supplementary Fig. 20. The TDB-treated perovskite film exhibits a narrower distribution than the control does, suggesting enhanced phase homogeneity across the WBG perovskite surface. In addition, the TDB-treated film has a mean QFLS value of 1.43 eV, whereas the control film has a lower mean QFLS value of 1.36 eV. These values agree well with the V_{oc} values of the corresponding devices

TDB-treated perovskite films after light soaking under 1-sun illumination for 20 min. The photoluminescence spectra were excited at 470 nm. **e**, Comparison of the J_{sc} of control and target WBG pero-SCs before and after 12 h of 1-sun light soaking, measured over a wide range of sweep rates (0.1 V s^{-1} to 1,000 V s^{-1}). **f**, Continuous MPP tracking of encapsulated control and target WBG pero-SCs in ambient air (around 30 °C and 60% relative humidity) under 1-sun illumination (ISOS-L-1).

(Fig. 2f) and suggest reduced non-radiative recombination after TDB treatment^{27–29}. We further investigated the contribution of interfacial energetics optimization to the QFLS improvement in the TDB-treated devices^{30,31}. We performed Kelvin probe force microscopy and ultraviolet photoelectron spectroscopy (UPS) measurements on the surfaces of pristine perovskite films and films treated with TDB. The TDB-treated film exhibited lower contact potential difference (CPD) values than the control did (Supplementary Fig. 21), indicating shallower work functions and a more *n*-type character at the top surface. On the basis of the UPS results (Supplementary Fig. 22), we constructed energy-level diagrams of the corresponding devices (Fig. 2g). The results reveal that treatment with TDB reduces the energy offset between the perovskite conduction band minimum and the C₆₀ lowest unoccupied molecular

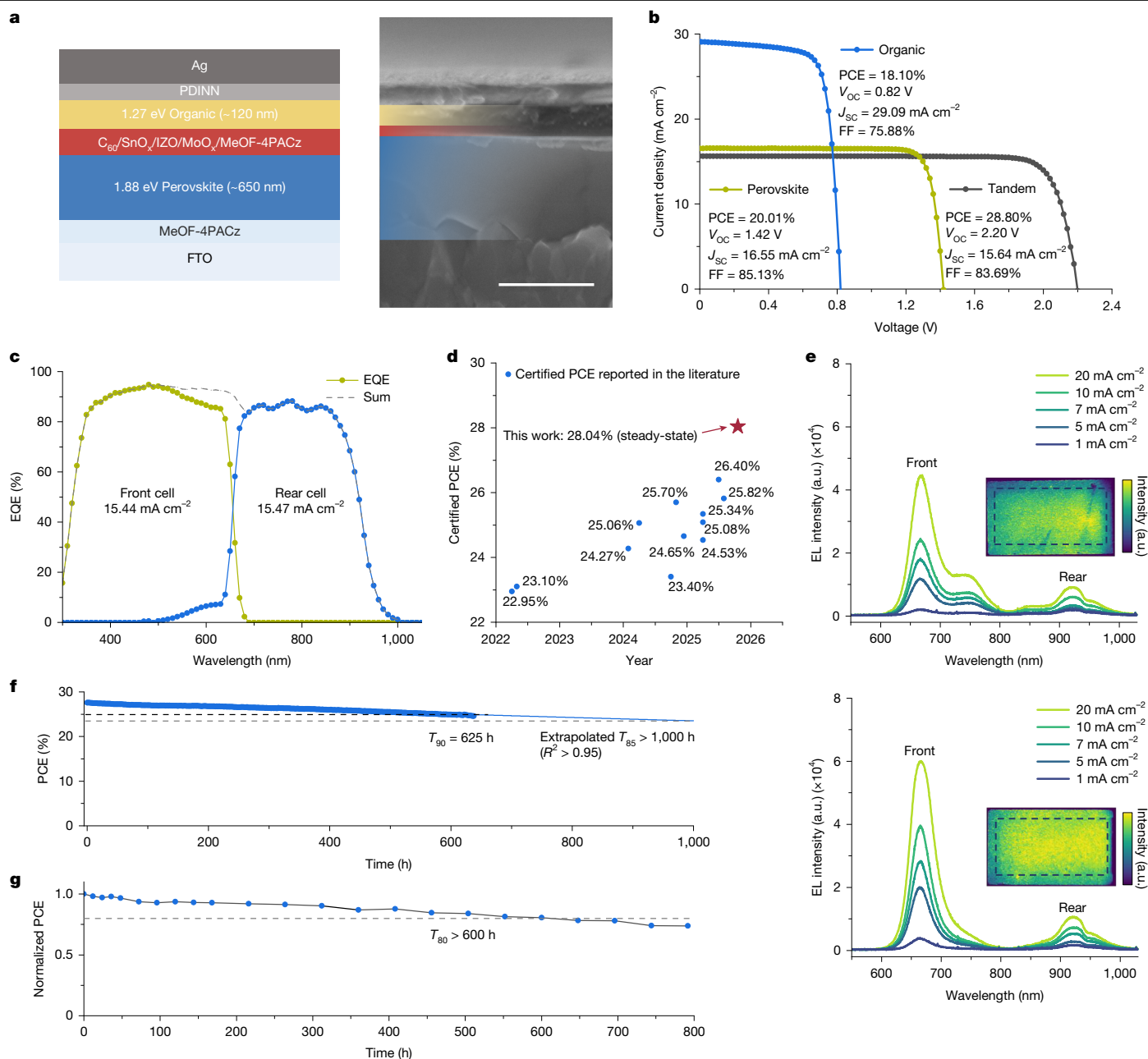


Fig. 4 | Photovoltaic performance and operational stability of perovskite-organic TSCs. **a**, Device architecture and cross-sectional SEM image of a perovskite-organic TSC. Scale bar, 500 nm. IZO, indium zinc oxide. **b**, J - V curves of a representative single-junction WBG pero-SC, single-junction OSC and perovskite-organic TSC. **c**, EQE spectra and integrated J_{sc} values of the front and rear cells in the representative perovskite-organic TSC. **d**, Summary of the certified efficiencies of perovskite-organic TSCs reported in the literature over the past three years. **e**, Electroluminescence (EL) spectra of aged control (top) and target (bottom) perovskite-organic TSCs measured at injection

current densities of 1, 5, 7, 10 and 20 mA cm $^{-2}$. Inset, electroluminescence image of the corresponding WBG perovskite front cell measured under an injection current density of 20 mA cm $^{-2}$; dashed box outlines the active area defined by sputtered IZO. **f**, Long-term MPP tracking of the encapsulated perovskite-organic TSCs in ambient air (around 30 °C and 60% relative humidity) under 1-sun illumination (ISOS-L-1). **g**, Thermal stability of encapsulated perovskite-organic TSCs under accelerated ageing at 85 °C in an N_2 atmosphere (ISOS-D-2).

orbital (LUMO), thereby optimizing the energy-level alignment at the perovskite- C_{60} interface, facilitating electron extraction and reducing interfacial energy losses^{30,31}. This effect might originate from interfacial dipoles introduced by TDB³². Furthermore, Mott-Schottky measurements (Fig. 2h) were performed to assess the built-in electric field (V_{bi}) of the pero-SCs, and the enhanced V_{bi} in the TDB-treated device was in agreement with the larger QFLS and the improved V_{oc} , which indicates enhanced driving force for carrier separation and reduction of charge recombination³³.

Photo-induced transformation of TDB

Building on the improved initial halide homogeneity achieved in stage I through the introduction of TDB, WBG perovskites still face the challenge of light-induced halide segregation in stage II, which undermines the operational stability of WBG pero-SCs^{1-3,19}. Defects reduce the activation barrier for halide-ion migration, thereby kinetically accelerating halide segregation^{3,6,18}. As well as enabling interactions with perovskite precursors during solution processing, the

trifluoromethyl diazirine moiety serves as a well-established photo-active group. This feature endows TDB with a photo-transformable nature, allowing it to transform into a new species under illumination with ultraviolet components. The photogenerated species inhibits the formation of iodide-related defects and suppresses phase segregation.

When illuminated with ultraviolet components, the trifluoromethyl diazirine group in TDB releases N_2 to generate a carbene intermediate^{34,35}. We measured the electron paramagnetic resonance (EPR) signals of the mixture of TDB and perovskite precursors under 1-sun simulated illumination containing ultraviolet, which indicated the generation of triplet carbene species (Supplementary Fig. 24). After exposure to oxygen, the triplet carbenes will capture oxygen^{36,37} and convert into a new species, TAB (Fig. 3a and Supplementary Fig. 25). We used ^{19}F NMR as an analytical tool to provide direct and quantitative evidence³⁸ for the TDB-to-TAB conversion in perovskite film (Supplementary Fig. 26).

Density functional theory (DFT) calculations indicate that TAB has a larger dipole moment (Fig. 3b) and adsorbs on the perovskite surface more strongly than TDB does (Fig. 3c and Supplementary Table 4), owing to its multisite anchoring mode involving electrostatic interactions through the ammonium group and coordination between the carbonyl oxygen and undercoordinated Pb sites. We synthesized TAB independently for subsequent characterization and analysis (Supplementary Figs. 27–30). X-ray photoelectron spectroscopy (XPS) measurements were performed to examine the interaction between TAB and the perovskite (Supplementary Fig. 31). TAB also increases the formation energies of iodide vacancies and iodide interstitials (Supplementary Fig. 32 and Supplementary Table 5). Transient photocurrent (TPC) measurements (Supplementary Fig. 33) were performed on the control, TDB-treated and TAB-treated pero-SCs. The TAB-treated device had the shortest decay time, suggesting reduced defect-assisted carrier trapping and charge accumulation^{4–6,39}. The reduced defect density can also suppress defect-mediated halide-ion migration⁵. We measured the temperature-dependent conductivity of the target and control samples and extracted the activation energy (E_a) for ion migration from the Arrhenius plot⁴⁰. As shown in Supplementary Fig. 34, the target sample exhibited a higher activation energy for ion migration.

Continuous-illumination photoluminescence measurements were performed to directly evaluate the role of TAB in suppressing phase segregation (Fig. 3d). The perovskite films were surface-modified by spin-coating TDB or TAB solutions (1 mg ml^{-1} in 2-propanol (IPA)) onto the perovskite surface. Whereas the control film developed a pronounced red-shifted emission at around 740 nm, assigned to an I-rich phase, the TAB-treated film showed almost no such peak throughout the measurement. Although TDB also showed some ability to suppress phase segregation, its effect was clearly less pronounced than that of TAB. This trend is further supported by the evolution of the I-rich-to-intrinsic peak ratio (Supplementary Fig. 35).

Furthermore, we compared the scan-rate-dependent J_{sc} of devices measured before and after ageing (Fig. 3e and Supplementary Figs. 36 and 37). For the fresh devices, the target devices showed weaker scan-rate dependence than the control devices, indicating reduced ionic losses in the as-fabricated devices, which is consistent with the role of TDB in promoting homogeneous halide mixing during film formation. After 12 h of ageing, this contrast became more pronounced, with the control devices exhibiting stronger scan-rate dependence, indicating larger ion-related losses that are consistent with increasing phase segregation during operation¹⁹. This behaviour further supports the synergistic roles of TDB in improving the initial halide homogeneity and TAB in suppressing defect-assisted halide segregation under illumination. The operational stability of encapsulated WBG pero-SCs was assessed under the ISOS-L-1 testing standard (Fig. 3f). The control device exhibited a reduction of around 25% in efficiency within 200 h,

but the target device maintained over 90% of its initial efficiency after 718 h of continuous 1-sun illumination containing ultraviolet in ambient air, indicating improved photostability. The relative changes in V_{oc}/J_{sc} and FF during MPP tracking are shown in Supplementary Fig. 38.

Photovoltaic performance and operational stability

On the basis of the reduced voltage loss and improved photostability of the target WBG pero-SCs, we constructed monolithic perovskite–organic TSCs incorporating a PM6:BTPSe-Ph4F:MO-IDIC-2F organic active layer^{41,42} (molecular structures are shown in Supplementary Fig. 40 and the EQE spectra of the single-junction OSC are in Supplementary Fig. 41). The tandem device architecture is fluorine-doped tin oxide (FTO)/MeOF-4PACz (ref. 43)/perovskite/ C_{60} /SnO_x/IZO/MoO_x/MeOF-4PACz/PM6:BTPSe-Ph4F:MO-IDIC-2F/PDINN/Ag (the device configuration and a cross-sectional SEM image of the TSC are shown in Fig. 4a). Detailed optimization of the perovskite active layers and TSCs (Supplementary Tables 6–8) confirmed that the 1.88-eV perovskite with a thickness of 650 nm is well matched with the OSC. The intrinsic stability analyses of the perovskite–organic TSCs are shown in Supplementary Figs. 42–44. The representative perovskite–organic TSC achieved a PCE of 28.80%, with a V_{oc} of 2.20 V, J_{sc} of 15.64 mA cm^{-2} and FF of 83.69% (Fig. 4b), and the PCE distribution of 30 devices is shown in Supplementary Fig. 45. The integrated J_{sc} values for the front and rear cells were 15.44 mA cm^{-2} and 15.47 mA cm^{-2} , respectively (Fig. 4c), which are consistent with the J_{sc} values from the J – V measurements. The device efficiency was further certified at the National Photovoltaic Industry Measurement and Testing Center (NPVM), with a steady-state PCE of 28.04% (Supplementary Fig. 46). Figure 4d and Supplementary Table 9 compare the certified steady-state PCE in this work with previously reported certified PCEs for perovskite–organic TSCs. The performance of larger-area devices is presented in Supplementary Fig. 47.

We measured the electroluminescence spectra of the control and target perovskite–organic TSCs after 240 h of ISOS-L-1 ageing to probe phase segregation in the WBG front cell (Fig. 4e). The aged control device exhibited an additional emission peak at around 740 nm across different injection current densities, which can be assigned to an I-rich phase. However, the target device retained a stable electroluminescence peak position, indicating greater resistance to bias-induced phase segregation even after light soaking⁴⁴. We kept the encapsulated devices in ambient air and used MPP tracking under simulated 1-sun illumination following the ISOS-L-1 protocol. The TSCs maintained 90% of their initial efficiency after about 625 h under continuous illumination. By fitting the degradation data with a biexponential decay model⁴⁵ ($R^2 > 0.95$), we extrapolated a projected T_{85} exceeding 1,000 h (Fig. 4f). Under accelerated ageing at 85 °C (ISOS-D-2), the encapsulated TSCs maintained over 80% of their initial performance after 600 h (Fig. 4g). The efficiency and stability of the devices in this study are a step towards the practical viability of perovskite–organic TSCs.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10869-x>.

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Methods

Materials

Fluorine-doped tin oxide (FTO) glass, formamidinium iodide (FAI), formamidinium bromide (FABr), methylammonium iodide (MAI), methylammonium bromide (MABr), lead iodide (PbI_2), lead bromide (PbBr_2) and nickel oxide (NiO_x) were purchased from Advanced Election Technology. Methylamine hydrochloride (MACl) and Ph-4PACz were obtained from Xi'an Yuri Solar. Rubidium iodide (RbI), caesium iodide (CsI), caesium bromide (CsBr), *N,N*-dimethylformamide (DMF), dimethyl sulfoxide (DMSO) and methanol (MeOH) were obtained from Sigma-Aldrich. IPA, ethyl acetate (EA), *o*-xylene and γ -butyrolactone (GBL) were purchased from J&K Scientific. PM6 was purchased from Solarmer Materials. The 1,8-diiodooctane was purchased from TCI. PDINN was purchased from Nanjing Zhiyan Technology and 4-[3-(trifluoromethyl)-3*H*-diazirin-3-yl]benzylammonium chloride was purchased from Aladdin. Silver was purchased from ZhongNuo Advanced Material (Beijing) Technology. *cis*-CyDAI₂ (ref. 27), MeOF-4PACz (ref. 43), 4-(trifluoroacetyl)benzylammonium chloride, BTPSe-Ph4F and MO-IDIC-2F were synthesized in our laboratory. Unless otherwise specified, all the chemicals and solvents were used without further purification. Tetrakis(dimethylamino)tin(IV) (TDMASn) and ultrapure water were purchased from MNT Micro and Nanotech.

Synthesis of MAPbI₃ powder

MAI (1.59 g, 10 mmol) and PbI_2 (4.61 g, 10 mmol) were added to a flask, followed by 15 ml GBL. The mixture was stirred at 70 °C until a clear precursor solution was obtained. Then, 200 ml of EA was placed in a beaker under stirring. The precursor solution was slowly added dropwise into the EA. After addition, a black precipitate gradually formed. The resulting precipitate was collected by centrifugation, and then washed with EA three times. Finally, the black powder was dried under vacuum at 60 °C for 12 h to obtain 5.20 g of MAPbI₃ powder (83.9% yield).

Synthesis of MAPbBr₃ powder

MABr (1.12 g, 10 mmol) and PbBr_2 (3.67 g, 10 mmol) were dissolved in a mixed solvent of 10 ml GBL and 2.5 ml DMSO with stirring at 70 °C until a clear precursor solution was obtained. The precursor solution was then slowly added dropwise into 200 ml EA under stirring, leading to the formation of an orange precipitate. The precipitate was collected by centrifugation, washed with EA three times and dried under vacuum at 60 °C for 12 h to obtain 4.10 g of MAPbBr₃ powder (85.6% yield).

Synthesis of 4-(trifluoroacetyl)benzylammonium chloride

Potassium (((tert-butoxycarbonyl)amino)methyl)trifluoroborate (622 mg, 2.62 mmol, 1.05 equivalents), 4'-chloro-2,2,2-trifluoroacetophenone (521 mg, 2.50 mmol, 1.00 equivalent), Pd(OAc)₂ (28.0 mg, 0.125 mmol, 0.05 equivalents), SPhos ligand (102.5 mg, 0.25 mmol, 0.1 equivalent) and K₂CO₃ (1035 mg, 7.5 mmol, 3.0 equivalents) were dissolved in toluene/H₂O (4:1, v/v; 20 ml/5 ml). The mixture was purged three times with argon and stirred at 85 °C for 22 h. After cooling to room temperature, the resulting mixture was extracted with EA. The organic layers were combined, dried over MgSO₄, filtered and concentrated in vacuo. The crude product was purified by column chromatography on silica gel using petroleum ether/EA (5:1) as the eluent. The product was collected and dried under a high vacuum overnight, providing the pure compound A (447 mg, 59%) as a white powder. A solution of HCl in cyclopentyl methyl ether (CPME; 3 M, 10.0 ml, 30.00 mmol) was added dropwise to a solution of the compound A (447 mg, 1.48 mmol) in dichloromethane at 0 °C. After the addition was complete, the reaction mixture was stirred at room temperature for 12 h. The solvent was removed in vacuo, and the residue was washed with ethyl acetate and subsequently dissolved in methanol. The methanol

solution was poured into diethyl ether to precipitate the product. The solvent was removed by rotary evaporation, and the resulting solid was washed several times with diethyl ether until it turned white. Finally, the white solid was collected by filtration and dried under vacuum, providing the pure compound 4-(trifluoroacetyl)benzylammonium chloride (282 mg, 79%) as a white powder.

Fabrication of WBG pero-SCs

The FTO glass substrates were sequentially cleaned by sonication with detergent, deionized water, acetone and IPA for 30 min, respectively. The cleaned FTO substrates were treated with UV-ozone for 30 min and transferred to a nitrogen-filled glove box immediately. A solution of MeOF-4PACz (0.5 mg ml⁻¹ in IPA) was ultrasonicated in an ice-water bath for 10 min before use and then spin-coated onto the FTO substrates at 4,000 rpm for 30 s. The resulting films were annealed at 120 °C for 10 min. To make FA_{0.75}Cs_{0.15}Rb_{0.10}Pb(I_{0.55}Br_{0.45})₃ perovskite precursors with a concentration of 1.6 M, CsI (41.6 mg), RbI (33.9 mg), CsBr (17.0 mg), FAI (123.8 mg), FABr (60.0 mg), PbI_2 (442.6 mg, 10 mol% excess) and PbBr_2 (293.6 mg) were dissolved in 1 ml mixed solvent of DMF:DMSO (v:v = 4:1), 10 mol% MACl was introduced into the precursor solution, and the mixture was shaken using a vortex mixer for 1–2 h until fully dissolved. The solution was filtered with a 0.22- μm PTFE filter before spin-coating. A solution of TDB (0.1 M) was prepared by mixing 25 mg TDB in 1 ml DMSO. Before spin-coating, the TDB solution was added to the perovskite precursor solution at the desired ratio. All perovskite precursor solutions were freshly prepared and used immediately before film deposition. The perovskite precursor solution (60 μl) was deposited onto the substrate and spin-coated at 4,000 rpm for 25 s. EA (200 μl) was dripped onto the perovskite film during the last 10 s of the spin-coating process. Alternatively, a gas-quench method could be used⁶. The temperature was maintained below 22 °C during device fabrication. The film was then annealed at 100 °C for 15 min. Next, spin-coating of *cis*-CyDAI₂ (0.25 mg ml⁻¹) was done onto the as-formed perovskite at 3,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. Afterwards, a 20-nm layer of C₆₀ was deposited by thermal evaporation, and 20 nm SnO_x was deposited subsequently by atomic layer deposition (ALD). The SnO_x was deposited by ALD with N₂ as the gas carrier. During ALD, the substrate temperature was maintained at 80 °C, the TDMASn precursor source was maintained at 70 °C and the H₂O source was maintained at room temperature. The dosing and purging time were 0.2 s and 10 s for TDMASn and 0.02 s and 15 s for H₂O. A 100-nm Ag electrode was deposited by thermal evaporation. For the stability test, the hole transport layer was replaced with NiO_x and Ph-4PACz. NiO_x aqueous ink (15 mg ml⁻¹) was spin-coated onto FTO glass at 2,500 rpm for 15 s. The resulting films were annealed at 150 °C for 15 min in ambient air and immediately transferred to a nitrogen-filled glove box. A solution of Ph-4PACz (0.5 mg ml⁻¹ in IPA) was coated by spin-coating at 4,000 rpm for 20 s and annealed at 100 °C for 10 min.

Fabrication of NBG OSCs

A 20-nm layer of MoO_x was thermally evaporated on top of indium tin oxide glass. Then, the substrates were transferred to a glove box filled with nitrogen. A solution of MeOF-4PACz (0.5 mg ml⁻¹ in IPA) was spin-coated at 4,000 rpm for 20 s and annealed at 100 °C for 5 min. PM6 was used as the donor, and BTPSe-Ph4F and MO-IDIC-2F were used as the acceptors. These materials were dissolved in *o*-xylene at a weight ratio of 1:1:0.2 with a total concentration of 20 mg ml⁻¹, and 0.3 vol% 1,8-diiodooctane was added as the solvent additive. After heating at 60 °C for 3 h, the solution was spin-coated onto the substrate at 3,000 rpm. The film was then annealed at 100 °C for 5 min. After this, 30 μl PDINN (1 mg ml⁻¹ in MeOH) was spin-coated onto the organic active layer at a spin speed of 3,000 rpm for 30 s. Finally, the film was transferred to the thermal evaporator system to deposit 100 nm Ag as the top electrode.

Fabrication of perovskite–organic TSCs

The front cell was prepared following the same procedure as that used for the single-junction pero-SC. After 20 nm C_{60} was deposited, 20 nm SnO_x was deposited by ALD. A 10-nm layer of IZO was sputtered onto the SnO_x layer by radio-frequency sputtering at 60 W. Subsequently, the process followed the same steps as those used in the preparation of a single-junction OSC. Finally, a 100-nm Ag electrode was deposited by thermal evaporation.

Characterization of solar cells

The J – V curves of the pero-SCs were measured by scanning the voltage from 1.45 V to -0.2 V for the reverse scan, and from -0.2 V to 1.45 V for the forward scan. A voltage scan from -0.2 to 1.0 V was applied for the OSCs. The voltage was scanned from 2.22 V to -0.2 V for the reverse scan of the TSCs, and from -0.2 V to 2.22 V for the forward scan of the TSCs. The voltage step in the scan was 20 mV with a delay time of 1 ms, using a Keysight B2901A Source Meter. The device areas were 0.092 cm^2 , as defined by metal aperture masks. The measurements were performed in an N_2 -filled glove box. An Enli-tech class AAA Solar Simulator (model SS-X180R-3A) with a 450-W xenon lamp was used as the light source. The light intensity was calibrated to 100 mW cm^{-2} using an SRC-2020 reference cell (traceable to NREL). The EQE responses were measured using the Solar Cell Spectral Response Measurement System QE-R3-011 (Enli Technology). The light intensity at each wavelength was calibrated using a standard single-crystal Si photovoltaic cell. When measuring the organic rear cells and perovskite front cells, 450-nm short-wave-pass filters and 800-nm long-wave-pass filters were used, respectively, together with the corresponding applied bias voltage (1.4 V bias voltage was applied during measurement of the organic rear cells and no bias voltage was applied during measurement of the perovskite front cells) to minimize artefacts arising from reverse bias of the saturated sub-cells.

Stability tests of solar cells

Stability tests were performed using a commercial test system (PAS-GV, Enli Technology). A white LED source covering the UV–visible spectral range was used to test the long-term stability of the perovskite–organic TSCs. The test was performed on encapsulated devices maintained in an ambient atmosphere (around 30°C , 60% relative humidity). MPP tracking was done using an LED solar simulator equivalent to AM1.5G, 100 mW cm^{-2} . During MPP tracking, the system measures an I – V curve and simultaneously determines V_{max} , which is then continuously applied to the device until the next I – V measurement provides new data. This process is repeated to achieve continuous MPP tracking. In addition, a light intensity detector is used to monitor the light intensity in real time. If the light intensity decreases, the output power is adjusted accordingly by the software. The synchronized light intensity data were also used to correct the stability measurements. For the thermal-ageing stability test, the encapsulated devices were stored on a hot plate at 85°C under a N_2 atmosphere in the dark. At predetermined time intervals, the devices were taken out for photovoltaic characterization, and the ageing process was resumed immediately after measurement. This interval-measurement procedure was repeated throughout the test to evaluate the high-temperature accelerated ageing stability of the devices. For humidity-ageing tests, the unencapsulated devices were placed in a humidity-controlled chamber maintained at 85% relative humidity. The devices were aged under dark conditions, and their photovoltaic performance was measured at regular intervals to evaluate moisture stability.

Solid-state ^{207}Pb NMR spectroscopy

Solid-state ^{207}Pb NMR spectra were recorded on a Bruker NEO 600 spectrometer using a Bruker 4.0-mm MAS NMR probe. The samples were spun at a spinning frequency of 5 kHz for magic-angle spinning with N_2 , using a Hahn-echo pulse sequence with a recycle delay of 1 s. The

resonance frequency was set to 125.68 MHz. MAPbI_3 and MAPbBr_3 (the synthesis procedures are provided above), together with TDB and TAB powders, were separately ground into fine, homogeneous powders. For the control sample, MAPbI_3 and MAPbBr_3 were mixed at a molar ratio of 5.5:4.5 (corresponding to the I/Br ratio of the perovskite composition used in this work) and vortexed for 5 min. For the TDB and TAB samples, MAPbI_3 , MAPbBr_3 and the respective additive were mixed at a molar ratio of 5.5:4.5:0.1 and vortexed for 5 min. All powder mixtures were heated in an 80°C oil bath under stirring, using round-bottom flasks of the same size, magnetic stir bars of the same size and the same stirring rate to ensure consistent heating and mixing conditions.

Other characterizations

The ^1H NMR spectra were measured using a Bruker Avance 400-MHz spectrometer. Fourier-transform infrared (FTIR) spectra were obtained with a Bruker VERTEX 70v spectrometer. In situ photoluminescence spectra were measured using the combination of a spectrometer (SpectraPro 300, Princeton Instruments) and a streak camera (C10910, Hamamatsu Photonics). Excitation light was generated using a fundamental pulse from a Ti:sapphire regenerative amplifier (Astrella, Coherent) and was then sent to an optical parametric amplifier (OPerA-Solo, Coherent) for the pump beam with a 470-nm wavelength. Appropriate long-pass and band-pass filters were used after the optical parametric amplifier (OPA) to ensure a clean spectrum. In situ photoluminescence spectra were recorded using the sequence acquisition in the focus mode of the streak camera. Software provided by Hamamatsu (HPD-TA, Hamamatsu Photonics) was used to collect and process all of the photoluminescence spectra. Photoluminescence mapping experiments were undertaken using a confocal laser scanning microscope (Leica STELLARIS 8) with a $40\times$ objective lens. Perovskite films were excited with a diode laser (470 nm), and the photoluminescence was collected from 600 nm to 700 nm. The steady-state photoluminescence signal was analysed by a Horiba Jobin Yvon system. Time-resolved photoluminescence profiles were obtained using an EPL (Edinburgh Instruments) with time-correlated single-photon counting capabilities. The films were excited by a 470-nm pulse laser with a repetition frequency of 20 kHz provided by a picosecond pulsed diode laser (FLS980, Edinburgh Instruments). The surface and cross-sectional morphology of the films and devices were investigated by SEM (Hitachi S-4800). For the cross-sectional image, the cross-sectional surface of the sample was coated with 1-nm-thick gold by sputtering to enhance the conductivity. AFM and Kelvin probe force microscopy measurements were performed using VistaScope from Molecular Vista, operated in dynamic mode using commercial gold-coated silicon cantilevers (NCHAU) from Nanosensors using tapping mode. In situ GIWAXS data were obtained at the 1W1A Diffuse X-ray Scattering Station, Beijing Synchrotron Radiation Facility (BSRF-1W1A). ToF-SIMS analysis was done on a ToF-SIMS M6 mass spectrometer (IONTOF). The samples for the GIWAXS measurements were prepared on FTO glass substrates using corresponding solutions of the samples. XRD patterns were obtained with an X-ray diffractometer (Malvern Panalytical) with $\text{Cu K}\alpha$ radiation at a scan rate of 2° min^{-1} . XPS was recorded on an ESCALA250XI instrument (Thermo Fisher Scientific) with an $\text{Al K}\alpha$ X-ray source. EPR measurements were performed using a BRUKER EMXplus-9.5/12. The sample was frozen in liquid nitrogen before measurement, and the spectra were recorded under in situ light irradiation. In the TPC and Mott–Schottky measurements, the pero-SCs were fabricated using the same method as mentioned above. The data were obtained by the all-in-one characterization platform, Paios (Fluxim AG). QFLS measurements were performed on samples in air using the QFLS-MAPER (manufactured by Enli Technology). The included software calculates the quantum yield and the implied open-circuit voltage (iV_{oc} /QFLS) of samples. Electroluminescence measurements were performed by Shenzhen Photon Quantum Lattice Technology. Temperature-dependent electrical

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conductivity was measured using a comprehensive physical property measurement system (PPMS DynaCool-9T).

Molecular simulations

DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP)^{46,47}. The generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) functional was used to account for exchange–correlation. The projector augmented-wave (PAW) potentials were used to describe the ionic cores and to explain the valence electrons using a plane-wave basis set with a kinetic energy cut-off of 450 eV. During structural optimization, the conjugate-gradient method was used and the convergence criteria for energy and force were set to 1×10^{-5} eV and 4×10^{-2} eV Å⁻¹, respectively. The Grimme's DFT-D3 correction was considered to describe the interactions between perovskite surface models and the adsorbed molecule⁴⁸. The Brillouin zone integration was performed using a $1 \times 1 \times 1$ *k*-point mesh. To simulate the defect surface of perovskite, a $3 \times 3 \times 2$ supercell with a vacuum-layer thickness of 15 Å in the *z* direction was constructed. The passivator was modelled as a monovalent ammonium cation, and a net charge of +1 e with a uniform compensating background was introduced in the calculations. The adsorption energies (E_{ads}) were calculated as $E_{\text{ads}} = E_{\text{slab/ad}} - E_{\text{slab}} - E_{\text{ad}}$, where $E_{\text{slab/ad}}$, E_{slab} and E_{ad} are the energy of the perovskite surface with the adsorbed molecule, the perovskite surface and the molecule, respectively. The defect formation energy is given as:

$$E_{\text{form}}[X^q] = E_{\text{tot}}[X^q] - E_{\text{tot}} + \sum_i n_i \mu_i + q(\varepsilon_{\text{VBM}} + E_{\text{F}}) + E_{\text{corr}}$$

Where E_{tot} and $E_{\text{tot}}[X^q]$ are the total energy of the pristine and the defective supercells, respectively; n_i and μ_i are the number and chemical potential of defective species, respectively; ε_{VBM} and E_{F} are the energy levels of the valence band maximum (VBM) and the Fermi level, respectively; and E_{corr} is a finite-size correction term.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors upon request. Source data are provided with this paper.

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Acknowledgements We thank the 1W1A-Diffuse X-ray Scattering Beamline of Beijing Synchrotron Radiation Facility for providing technical support and assistance in GIWAXS data collection; Enli Technology for QFLS measurement equipment; Shenzhen HUASUAN Technology for assistance with theoretical calculations; and N. Wu for assistance with solid-state ²⁰⁷Pb NMR measurements and data analysis.

Author contributions L.M. and R.W. conceived the idea and designed the experiments. R.W. performed device characterization and analysed data. R.W., S.Q., T.Z. and S.W. participated in the fabrication and characterization of tandem solar cells. S.Z. and X.J. participated in device fabrication. S.Q. and H.L. performed DFT calculations and analysis. Y.W., S.L., M.L. and Y.F. participated in the photoluminescence and optoelectronic property analyses. Y.G., H.H. and P.L. synthesized the organic photovoltaic materials. Y.C. participated in the design and analysis of in situ GIWAXS experiments. J.Z. performed in situ photoluminescence measurements. X.L. participated in the data analysis and discussion. L.M. and Y.L. supervised the project. R.W., L.M. and Y.L. wrote the manuscript. All authors contributed to the work.

Funding The authors disclose support for the research of this work from the National Key Research and Development Program of China (2024YFB4205200), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDB0520102), the National Natural Science Foundation of China (52173188), and the Key R&D and Achievement Transformation Plan Project of Inner Mongolia Autonomous Region (2025YFHH0021).

Competing interests The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10869-x>.

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Peer review information Nature thanks the anonymous reviewers for their contribution to the peer review of this work. Peer reviewer reports are available.

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

Area of the tested solar cells is 0.1225 cm² defined by optical microscope (Olympus BX51).

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Device area is determined by a mask of 0.092 cm².

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

We tested current density-voltage (J-V) plots in both forward and backward direction

Voltage scan conditions

☒ Yes
☐ No

In forward direction, the voltage was scanned from 2.22 V to -0.2 V. In reverse direction, the voltage was scanned from -0.2 V to 2.22 V. The voltage step and delay time were 20 mV and 1 ms, respectively. Relative information is provided in method section.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Devices were characterized at room temperature in In N₂-filled glove box.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

No preconditioning protocol.

Explain why this information is not reported/not relevant.

Stability of the J-V characteristic

☒ Yes
☐ No

Relative information is provided in text.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Hysteresis is small and the related data is included in the manuscript.

Explain why this information is not reported/not relevant.

Related experimental data

☐ Yes
☒ No

Provide a description of the related experimental data.

We didn't find the unusual behaviour.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

The data is included in the manuscript. We show the EQE measurement details in the Methods section.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The data is included in the manuscript. We show the EQE measurement details in the Methods section. <i>Explain why this information is not reported/not relevant.</i>
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	The device is tandem solar cell. The bias illumination obtained by 450-nm short-wave-pass filters and 800-nm long-wave-pass filters were used for the measurements of the rear subcells and front subcells, together with the corresponding applied bias voltage.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Relative information is provided in method section. <i>Explain why this information is not reported/not relevant.</i>
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Relative information is provided in method section. <i>Explain why this information is not reported/not relevant.</i>
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Relative information is provided in method section. <i>Explain why this information is not reported/not relevant.</i>
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Device area is determined by a mask with area of 0.0918 cm ² . <i>Explain why this information is not reported/not relevant.</i>
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	The variation is within 0.3% <i>Explain why this information is not reported/not relevant.</i>
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	The photovoltaic performance of our devices was confirmed by National Photovoltaic Industry Measurement and Testing Center. <i>Explain why this information is not reported/not relevant.</i>
A copy of any certificate(s)	☒ Yes ☐ No	A copy is provided in Supplementary Information. <i>Explain why this information is not reported/not relevant.</i>
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	The average PCE of the single-junction solar cells is obtained from more than 20 independent devices. <i>Explain why this information is not reported/not relevant.</i>
Statistical analysis of the device performance	☒ Yes ☐ No	We have given statistical data of device performance. <i>Explain why this information is not reported/not relevant.</i>
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	For the stability tests, all the device were encapsulated. The operational stability measurement was performed under a LED that was adjusted to 1 sun in ambient atmosphere. <i>Explain why this information is not reported/not relevant.</i>

